



FASR Science Case # WG1-5

Diagnosing the Hot Onset of Large Solar Flares: Coronal Magnetic Fields and the Thermal-to-Nonthermal Transition

Milo Buitrago-Casas¹

¹Space Sciences Lab - UC Berkeley

1 Science Goal(s)

The goal is to determine when and where an accelerated electron population first appears during the hot onset of a flare. FASR will follow the early thermal source, measure its microwave spectrum and magnetic-field constraints, and either detect a nonthermal excess or place a quantitative upper limit. The decisive observable is the first statistically secure departure from a thermal free-free/gyro model, measured relative to the evolving source and tied to the impulsive time history.

2 Scientific Rationale

2.1 Scientific Importance

Hot X-ray onsets show that flare plasma can reach roughly 10–15 MK before hard X-rays become detectable (Hudson et al. 2021; Battaglia et al. 2023). That sequence does not establish that accelerated electrons are absent; it sets an observational upper limit. The scientific question is whether acceleration begins simultaneously with the hot source, follows it after a measurable delay, or is hidden by the changing thermal spectrum.

Microwave imaging spectroscopy adds a direct test because thermal free-free and gyro emission have different spectral and polarization behavior from nonthermal gyrosynchrotron emission. Kou et al. (2025) demonstrated precursor microwave spectroscopy with EOVS, finding a hot, dense source before the impulsive phase and requiring a nonthermal component later. FASR should extend that test to the first detectable excess, with enough spatial and temporal fidelity to distinguish a real onset from changing source size, opacity, or calibration.

The magnetic measurement is a constraint on the interpretation, not a stand-in for the reconnection guide field. Compare the thermal and nonthermal evolution with families of magnetic geometries and retain the uncertainty in viewing angle, source depth, and unresolved structure. A population of detections and nondetections is needed before any claim about prediction of flare magnitude can be made.

2.2 Uniqueness to FASR Capabilities

X-ray and EUV data constrain hot plasma and energetic electrons, while magnetograms constrain the photospheric boundary. FASR uniquely supplies spatially registered, broadband microwave spectra and Stokes V through the buildup and impulsive transition. It can fit thermal and nonthermal components together and locate the first excess relative to the hot source. EOVS is a valuable pathfinder for the analysis, but its observing cadence and array sampling must not be turned into FASR requirements.

2.3 Synergies

GOES and imaging X-ray spectroscopy provide the thermal time history and hard-X-ray comparison. EUV spectroscopy constrains density, temperature, and flows. DKIST and synoptic magnetograms provide boundary fields and geometry. The onset time must be compared across these diagnostics using their detection thresholds, rather than treating the first detection in any one band as the physical onset.

2.4 Measurements Required by FASR

Obtain Stokes I and V images before and through flare onset. Fit each time step with competing thermal-only and thermal-plus-nonthermal models. The required product is a map of the nonthermal excess, its uncertainty, its spectrum, and an upper limit when no excess is recovered. The timing requirement is set by the shortest interval over which the competing onset hypotheses differ in the injection tests.

The author-submitted broadband concept is retained as a starting point. Coverage, spectral sampling, image fidelity, and sensitivity must be chosen by recovering injected excesses at a stated completeness and false-detection rate. No fixed channel width, dynamic range, or brightness rms is adopted here without that test.

III. Science Requirements Tables

(A) Observational Target Description

The field must contain the preflare hot source, its active-region background, and the later impulsive structures. Smaller events and candidates without a later large flare are useful controls. The relevant brightness is the change between competing thermal and nonthermal models, not the plasma temperature itself. Quantitative sensitivity targets are therefore reported as recovery criteria until source-injection tests supply their numerical values.

(A) OBSERVATIONAL TARGET	
Type of observation (what defines a ‘target’)	Active regions observed before and through flare onset. Measure thermal evolution and search for the first nonthermal microwave excess in Stokes I and V.

(A) OBSERVATIONAL TARGET (continued)		
Number of targets	Baseline: one focused event at a time. Goal: a sample covering flare classes and events with and without a later large flare.	
Size of a single target (arcsec x arcsec)	A few arcminutes, including the hot source and surrounding active-region emission.	
Distribution of all targets (arcmin x arcmin)	Full-disk monitoring for target selection; focused imaging on the selected active region.	
Peak brightness (sfu/beam or Kelvin)	Source dependent. Use thermal-source and injected nonthermal brightness distributions from observed events.	
RMS brightness (sfu/beam or Kelvin)	Baseline: recover the minimum injected nonthermal excess at $\geq 5\sigma$ with the required completeness; numerical rms is a validation output.	
Expected circularly polarized flux density	Stokes V (sfu/beam)	Baseline: retain signed V; the signal is model dependent and must include thermal and nonthermal contributions.
	V/I	Use the injected source distribution; do not impose a universal percentage.
Expected linearly polarized flux density	Stokes Q or U (sfu/beam)	Deferred for this case; Q/U is not needed for the first-excess test.
	Q/I or U/I	Deferred.

(B) Spectro-Temporal Requirements Discussion

The first excess can develop during the 5–10 minute buildup described in the submitted case, so a few-second cadence is scientifically justified for the focused sequence. Retain 5 s as the baseline only for intervals in which the onset hypotheses differ on that timescale. A 1 s goal is useful for impulsive transitions if it does not compromise the required spectral coverage and sensitivity. Slower full-disk monitoring can trigger the focused mode.

The spectrum must sample the curvature and turnover that separate thermal and nonthermal fits. Set channel spacing as a fraction of frequency and choose the number of channels from the injection test. A baseline $\Delta\nu/\nu \leq 2\%$ and a goal of $\leq 1\%$ are science-driven starting points; a fixed 100 MHz channel is not required. The full diagnostic spectrum must be available within each 5 s baseline interval so that evolution is not confused with spectral shape.

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	Author-submitted broadband starting range; validate the useful interval from thermal/nonthermal model recovery.

(B) SPECTRAL-TEMPORAL REQUIREMENTS (continued)	
Instantaneous Bandwidth (GHz/pol)	Baseline: enough tunable coverage to distinguish the competing spectra within 5 s; implementation deferred.
Spectral resolution [MHz]	Baseline: $\Delta\nu/\nu \leq 2\%$. Goal: $\leq 1\%$ where the injection test shows a turnover or curvature requires it.
Temporal resolution (in seconds)	Baseline: 5 s during the hot-onset sequence. Goal: 1 s for the impulsive transition if recovery tests support it.

(C) Polarization Product Discussion

Stokes I is required to fit the thermal and nonthermal components. Stokes V is baseline because circular polarization provides an independent constraint on the magnetic field and emission mechanism. Q and U are deferred because they are not needed for the first-excess discrimination and may be reduced by propagation and unresolved structure.

(C) POLARIZATION DATA PRODUCTS REQUIRED	
y	Stokes I
n	Stokes Q
n	Stokes U
y	Stokes V

(D) Imaging Requirements Discussion

The image must separate the hot source from the active-region background and track its position through onset. Angular resolution and field size are set by the source-injection population: the simulation must include compact sources, diffuse loops, source growth, and partially resolved components. Frequency-dependent beams may be modeled explicitly; a fixed common beam is not a science requirement. Dynamic range and sensitivity are accepted only when injected nonthermal excesses are recovered with the required completeness and false-detection rate.

(D) IMAGING REQUIREMENTS	
Required angular resolution (arcsec) (single value or range)	Baseline: resolve the smallest source separation that changes the thermal/nonthermal model fit; determine numerically from injection tests.
Largest angular scale required (arcsec)	Baseline: the active-region source and surrounding background; determine from the event population.
Mapped image size (arcmin x arcmin)	Baseline: a few arcminutes for the focused field; full-disk monitoring for event selection.

(D) IMAGING REQUIREMENTS (continued)		
Required pixel resolution (arcsec)	Baseline: at least three pixels across the delivered beam.	
Number of output/image channels	Baseline: set by recovery of the thermal/nonthermal spectral difference; no fixed channel count.	
Output bandwidth (minimum and maximum frequency - GHz)	Author-submitted broadband starting range; validate the useful interval from model recovery.	
Channel width (MHz)	Baseline: fractional width $\leq 2\%$; goal $\leq 1\%$ where needed by the onset test.	
Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each)	Baseline: $\geq 5\sigma$ recovery of the faintest injected nonthermal excess in I and V at the required completeness; numerical rms is a validation output.	
Dynamic range within image (if polarization products required define for each)	Baseline: no false nonthermal detection from residuals around the thermal source; numerical ratio is set by injection tests.	
Polarization accuracy (%)	Baseline: I-to-V leakage below the weakest V signature used in the onset test; determine from injected calibration-error trials.	
Zero spacing/total power required?	Baseline: retain the absolute and extended emission needed to fit the thermal component; implementation is a design validation item.	
Required maximum latency (in seconds, or N/A)	N/A for retrospective onset timing; operational alert latency is a separate requirement.	
Required flux density scale calibration accuracy		Baseline: preserve spectral changes used in the thermal/nonthermal fit.
	Goal	Set by the change in detection completeness in injection tests.
	Stretch	Absolute scale adequate for cross-instrument energy comparisons.
		Deferred until the source population is defined.

2.5 Other Performance or Functional Requirements

Retain pre-onset nondetections, calibration uncertainties, timestamps, pointing, beam metadata, and the Stokes sign convention. The analysis must publish upper limits when a nonthermal component is not recovered. A later statistical study can compare hot-onset properties with flare magnitude; that predictive claim is outside the single-event measurement requirement.

3 Appendix

The validation experiment begins with thermal source models constrained by X-ray and EUV observations. Inject nonthermal populations with a range of onset times, source locations, spectra, filling factors, and strengths. Render thermal-only and thermal-plus-nonthermal data through candidate frequency sampling, spatial response, polarization leakage, calibration errors, and time cadence. Fit both models without using the injection label. The requirements are accepted when the minimum excess is recovered with a stated completeness and false-detection rate and when nondetections produce useful upper limits.

Kou et al. (2025) is an observational pathfinder for precursor microwave spectroscopy; its EOVS setup and integration history do not define FASR requirements. The 5 s baseline is retained because the scientific question concerns onset timing on the buildup-to-impulsive transition. The cadence should be relaxed or tightened after the event population and recovery curves are measured.

Reference

- Hudson, H. S., Simões, P. J. A., Fletcher, L., Hayes, L. A., and Hannah, I. G. (2021). Hot X-ray onsets of solar flares. *MNRAS*, 501, 1273–1281. doi:10.1093/mnras/staa3664.
- Battaglia, A. F., et al. (2023). The existence of hot X-ray onsets in solar flares. *A&A*, 679, A139. doi:10.1051/0004-6361/202347706.
- Arnold, H., et al. (2021). Electron Acceleration during Macroscale Magnetic Reconnection. *Physical Review Letters*, 126, 135101. doi:10.1103/PhysRevLett.126.135101.
- Dahlin, J. T., Drake, J. F., and Swisdak, M. (2017). The role of three-dimensional transport in driving enhanced electron acceleration during magnetic reconnection. *Physics of Plasmas*, 24, 092110. doi:10.1063/1.4986211.
- Fleishman, G. D., Nita, G. M., and Gary, D. E. (2015). Energy partitions and evolution in a purely thermal solar flare. *ApJ*, 802, 122. doi:10.1088/0004-637X/802/2/122.
- Kou, Y., Cheng, X., Yu, S., Luo, Y., Chen, B., and Ding, M. (2025). Microwave Imaging Spectroscopy Diagnosis of the Slow-rise Precursor of a Major Solar Eruption. *ApJL*, 988, L62. doi:10.3847/2041-8213/adf063.